Diag Solutions

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1. **1.** If $\tan A = \frac{4}{3}$, find the value of $\sin A + \cos A$.

   Answer:

   $$\frac{7}{5}$$

   Solution:

   1. Use the definition of $\tan A = \frac{\text{opposite}}{\text{adjacent}}$ to find the hypotenuse using the Pythagorean theorem.
   2. Calculate $\sin A$ and $\cos A$ and find their sum.

   Derivation:

   $$\tan A = \frac{BC}{AB} = \frac{4}{3}$$
   $$\implies BC = 4k, AB = 3k.AC = \sqrt{(4k)^2 + (3k)^2} = 5k.\sin A = \frac{4}{5}, \cos A = \frac{3}{5}.\sin A + \cos A = \frac{4}{5} + \frac{3}{5} = \frac{7}{5}$$

   Introduction to Trigonometry — Trigonometric Ratios

2. **2.** Evaluate: $\frac{2\tan^2 45° + \cos^2 30° - \sin^2 60°}{1}$

   Answer:

   $$2$$

   Solution:

   1. Substitute the standard trigonometric values for $45°, 30°$, and $60°$.
   2. Simplify the expression using arithmetic operations.

   Derivation:

   $$\tan 45° = 1, \cos 30° = \frac{\sqrt{3}}{2}, \sin 60° = \frac{\sqrt{3}}{2}.\text{Expression} = 2(1)^2 + (\frac{\sqrt{3}}{2})^2 - (\frac{\sqrt{3}}{2})^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2$$

   Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

3. **3.** If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0° < A + B \leq 90°$ and $A > B$, find A and B.

   Answer:

   $$A = 45°, B = 15°$$

   Solution:

1. Use the inverse trigonometric values to set up a system of linear equations.
2. Solve the equations for A and B.

Derivation:

$$A - B = 30^\circ \text{and A} + B = 60^\circ.(A - B) + (A + B) = 30^\circ + 60^\circ$$
$$\implies 2A = 90^\circ$$
$$\implies A = 45^\circ.B = 60^\circ - 45^\circ = 15^\circ$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

4. **4.** Express $\cos A$ in terms of $\tan A$.

Answer:

$$\cos A = \frac{1}{\sqrt{1 + \tan^2 A}}$$

Solution:

1. Use the identity $\sec^2 A = 1 + \tan^2 A$.
2. Relate $\sec A$ to $\cos A$ using the reciprocal relation.

Derivation:

$$\sec^2 A = 1 + \tan^2 A$$
$$\implies \sec A = \sqrt{1 + \tan^2 A}.\text{Since } \cos A = \frac{1}{\sec A}, \cos A = \frac{1}{\sqrt{1 + \tan^2 A}}$$

Introduction to Trigonometry — Trigonometric Identities

5. **5.** Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

Answer:

$$1$$

Solution:

1. Apply the complementary angle relation $\tan \theta = \cot(90^\circ - \theta)$.
2. Simplify the fraction.

Derivation:

$$\tan 65^\circ = \tan(90^\circ - 25^\circ) = \cot 25^\circ.\text{Expression} = \frac{\cot 25^\circ}{\cot 25^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

6. **6.** If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B.

Answer:

$$A = 45^\circ, B = 15^\circ$$

Solution:

1. Use the values of tangent from the trigonometric table for $A + B$ and $A - B$.
2. Set up a system of linear equations in A and B.
3. Solve the system using elimination method.

Derivation:

$$\begin{aligned}
\tan(A + B) &= \sqrt{3} \\
\implies A + B &= 60^\circ \quad (1) \\
\tan(A - B) &= \frac{1}{\sqrt{3}} \\
\implies A - B &= 30^\circ \quad (2) \\
(1) + (2) : 2A &= 90^\circ \\
\implies A &= 45^\circ \\
(1) - (2) : 2B &= 30^\circ \\
\implies B &= 15^\circ
\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

7. **7.** If $3\cot\theta = 2$, evaluate the value of $\frac{4\sin\theta - 3\cos\theta}{2\sin\theta + 6\cos\theta}$.
Answer:

$$\frac{1}{3}$$

Solution:

1. Given $3\cot\theta = 2$, find $\cot\theta = \frac{2}{3}$.
2. Divide numerator and denominator by $\sin\theta$ to express in terms of $\cot\theta$.
3. Substitute the value of $\cot\theta$ and simplify.

Derivation:

$$\begin{aligned}
\frac{4\sin\theta - 3\cos\theta}{2\sin\theta + 6\cos\theta} &= \frac{4 - 3\cot\theta}{2 + 6\cot\theta} \\
&= \frac{4 - 3(2/3)}{2 + 6(2/3)} \\
&= \frac{4 - 2}{2 + 4} = \frac{2}{6} = \frac{1}{3}
\end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

8. **8.** Prove that $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.
Answer:

Proof completed.

Solution:

1. Expand both squares using $(a+b)^2 = a^2 + b^2 + 2ab$.
2. Group $\sin^2 A + \cos^2 A = 1$ and use identities $\csc^2 A = 1 + \cot^2 A$ and $\sec^2 A = 1 + \tan^2 A$.
3. Simplify the product terms $2 \sin A \csc A$ and $2 \cos A \sec A$ to 2.

Derivation:

$$\begin{aligned} LHS &= \sin^2 A + \csc^2 A + 2 + \cos^2 A + \sec^2 A + 2 \\ &= (\sin^2 A + \cos^2 A) + 4 + (1 + \cot^2 A) + (1 + \tan^2 A) \\ &= 1 + 4 + 1 + 1 + \tan^2 A + \cot^2 A = 7 + \tan^2 A + \cot^2 A = RHS \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

9. **9.** If $\sin\theta + \sin^2\theta = 1$, prove that $\cos^2\theta + \cos^4\theta = 1$.

Answer:

Proof completed.

Solution:

1. From $\sin\theta + \sin^2\theta = 1$, express $\sin\theta = 1 - \sin^2\theta$.
2. Substitute $\sin\theta = \cos^2\theta$.
3. Square both sides to get $\sin^2\theta = \cos^4\theta$ and substitute into the identity.

Derivation:

$$\begin{aligned} \sin\theta &= 1 - \sin^2\theta \\ \implies \sin\theta &= \cos^2\theta \\ \text{Square both sides: } \sin^2\theta &= \cos^4\theta \\ \text{Substitute into } \cos^2\theta + \cos^4\theta : \sin\theta + \sin^2\theta &= 1 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

10. **10.** Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$.

Answer:

$$\frac{3\sqrt{2} - \sqrt{6}}{8}$$

Solution:

1. Substitute the standard trigonometric values: $\cos 45^\circ = 1/\sqrt{2}$, $\sec 30^\circ = 2/\sqrt{3}$, $\csc 30^\circ = 2$.
2. Simplify the denominator.
3. Rationalize the resulting expression.

Derivation:

$$
\begin{aligned}
\frac{1/\sqrt{2}}{2/\sqrt{3}+2} = \frac{1/\sqrt{2}}{(2+2\sqrt{3})/\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{2}(2+2\sqrt{3})} \\
= \frac{\sqrt{3}}{2\sqrt{2}+2\sqrt{6}} = \frac{\sqrt{3}(2\sqrt{6}-2\sqrt{2})}{4(6-2)} = \frac{6\sqrt{2}-2\sqrt{6}}{16} = \frac{3\sqrt{2}-\sqrt{6}}{8}
\end{aligned}
$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

11. 11. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$.

Answer:

$$\csc A + \cot A$$

Solution:

1. Divide numerator and denominator by $\sin A$.
2. Rewrite the expression in terms of $\cot A$ and $\csc A$.
3. Replace 1 in the numerator with $\csc^2 A - \cot^2 A$.
4. Factorize the numerator using $a^2 - b^2 = (a-b)(a+b)$.
5. Cancel the common term to simplify.

Derivation:

$$\frac{\cot A - 1 + \csc A}{\cot A + 1 - \csc A} = \frac{\cot A + \csc A - (\csc^2 A - \cot^2 A)}{\cot A - \csc A + 1} = \frac{(\cot A + \csc A) - (\csc A - \cot A)(\csc A + \cot A)}{\cot A - \csc A + 1} = \frac{(\cot A + \csc A)(1 - (\csc A - \cot A))}{\cot A - \csc A + 1} = \cot A + \csc A$$

Introduction to Trigonometry — Trigonometric Identities

12. 12. If $\sin\theta + \cos\theta = \sqrt{3}$, then prove that $\tan\theta + \cot\theta = 1$.

Answer:

$$\tan\theta + \cot\theta = 1$$

Solution:

1. Square both sides of the given equation.
2. Expand $(\sin\theta + \cos\theta)^2$ to get $\sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 3$.
3. Use identity $\sin^2\theta + \cos^2\theta = 1$ to find $2\sin\theta\cos\theta = 2$, so $\sin\theta\cos\theta = 1$.
4. Express $\tan\theta + \cot\theta$ as $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$.
5. Simplify the fraction to $\frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{1} = 1$.

Derivation:

$$
\begin{aligned}
(\sin\theta + \cos\theta)^2 = 3 \\
\implies 1 + 2\sin\theta\cos\theta = 3 \\
\implies \sin\theta\cos\theta = 1; \quad \tan\theta + \cot\theta = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} = 1
\end{aligned}
$$

13. **13.** Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

Answer:

$$\frac{67}{12}$$

Solution:

1. State values: $\cos 60^\circ = 1/2$, $\sec 30^\circ = 2/\sqrt{3}$, $\tan 45^\circ = 1$, $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$.
2. Substitute values into the expression.
3. Calculate squares: $\cos^2 60^\circ = 1/4$, $\sec^2 30^\circ = 4/3$, $\tan^2 45^\circ = 1$.
4. Simplify the numerator: $5(1/4) + 4(4/3) - 1$.
5. Simplify the denominator: $(1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 = 1$.
6. Final calculation: $5/4 + 16/3 - 1 = (15 + 64 - 12)/12 = 67/12$.

Derivation:

$$\frac{5(1/4) + 4(4/3) - 1}{1} = \frac{5}{4} + \frac{16}{3} - 1 = \frac{15 + 64 - 12}{12} = \frac{67}{12}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

14. **14.** If $\sec\theta = x + \frac{1}{4x}$, prove that $\sec\theta + \tan\theta = 2x$ or $\frac{1}{2x}$.

Answer:

$$\sec\theta + \tan\theta = 2x \text{ or } \frac{1}{2x}$$

Solution:

1. Use the identity $\tan^2\theta = \sec^2\theta - 1$.
2. Substitute $\sec\theta = x + \frac{1}{4x}$.
3. Calculate $\tan^2\theta = (x + \frac{1}{4x})^2 - 1$.
4. Simplify to $\tan^2\theta = x^2 + \frac{1}{16x^2} + \frac{1}{2} - 1 = x^2 + \frac{1}{16x^2} - \frac{1}{2}$.
5. Notice this is a perfect square: $(x - \frac{1}{4x})^2$.
6. Thus $\tan\theta = \pm(x - \frac{1}{4x})$.
7. Sum $\sec\theta + \tan\theta$ for both cases.

Derivation:

$$\tan^2\theta = (x + \frac{1}{4x})^2 - 1 = x^2 + \frac{1}{16x^2} - \frac{1}{2} = (x - \frac{1}{4x})^2$$
$$\implies \tan\theta = \pm(x - \frac{1}{4x}); \quad \text{Case 1: } (x + \frac{1}{4x}) + (x - \frac{1}{4x}) = 2x; \quad \text{Case 2: } (x + \frac{1}{4x}) - (x - \frac{1}{4x}) = \frac{1}{2x}$$

Introduction to Trigonometry — Trigonometric Identities

15. **15.** To draw a pair of tangents to a circle which are inclined to each other at an angle of $60°$, it is required to draw tangents at the endpoints of those two radii of the circle, the angle between which is:

Answer: (C)

Solution:

1. The angle between the tangents and the angle between the radii at the points of contact are supplementary.
2. Let the angle between the radii be θ.
3. $\theta + 60° = 180°$
4. $\theta = 120°$

Derivation:

$$\theta = 180° - 60° = 120°$$

Constructions — Construction of tangents

16. **16.** If a point P is 10 cm away from the centre of a circle of radius 6 cm, the length of the tangent drawn from P to the circle is:

Answer: (B)

Solution:

1. The radius is perpendicular to the tangent at the point of contact.
2. This forms a right-angled triangle with the hypotenuse as the distance from the centre.
3. Use Pythagoras theorem: $d^2 = r^2 + t^2$
4. $10^2 = 6^2 + t^2$

Derivation:

$$t = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ cm}$$

Constructions — Construction of tangents

17. **17.** How many tangents can be drawn from an external point to a circle?

Answer: (B)

Solution:

1. A tangent touches the circle at exactly one point.
2. From a point outside a circle, exactly two distinct tangents can be drawn to the circle.

Derivation:

$$\text{Number of tangents} = 2$$

18. **18.** To divide a line segment AB in the ratio $3 : 4$, we draw a ray AX such that $\angle BAX$ is acute. Then we mark points on AX at equal distances. The minimum number of points to be marked is:

Answer: (C)

Solution:

1. To divide a line segment in ratio $m : n$, we need to mark $m + n$ points on the ray.
2. Here $m = 3$ and $n = 4$.
3. Total points $= 3 + 4 = 7$

Derivation:

$$3 + 4 = 7$$

Constructions — Construction of tangents

19. **19.** The angle between a tangent to a circle and the radius passing through the point of contact is:

Answer: (C)

Solution:

1. By the theorem of tangents, the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Perpendicular means 90°.

Derivation:

$$\text{Angle} = 90^\circ$$

Constructions — Construction of tangents

20. **20.** Assertion (A): The length of the tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Answer: (D)

Solution:

1. The length of a tangent is independent of the radius length.
2. The theorem states that a tangent is perpendicular to the radius at the point of contact.

Derivation:

$$A : \text{False}, R : \text{True}$$

21. **21.** Assertion (A): To construct a pair of tangents to a circle from an external point, the angle between the tangents is supplementary to the angle subtended by the line segment joining the points of contact at the center. Reason (R): The sum of opposite angles in a cyclic quadrilateral is 180°.

Answer: (A)

Solution:

1. The quadrilateral formed by the center, two contact points, and the external point has two right angles.
2. The sum of angles is 360°, so the remaining two must sum to 180°.

Derivation: A: True, R: True, R explains A *Constructions — construction of tangents*

22. **22.** Assertion (A): It is impossible to construct a tangent to a circle from a point lying inside the circle. Reason (R): A tangent touches the circle at only one point.

Answer: (A)

Solution:

1. A line passing through an interior point will always be a secant, not a tangent.
2. A tangent by definition meets the circle at exactly one point on the circumference.

Derivation: A: True, R: True, R explains A *Constructions — construction of tangents*

23. **23.** Assertion (A): The number of tangents that can be drawn from a point on the circle to the circle is exactly two. Reason (R): A point on the circle lies on the circumference.

Answer: (D)

Solution:

1. Only one tangent can be drawn from a point on the circle.
2. The reason is a true statement about the location of the point.

Derivation:

$$A : \text{False}, R : \text{True}$$

Constructions — construction of tangents

24. **24.** Assertion (A): When constructing tangents to a circle from an external point P, we bisect the line segment OP where O is the center. Reason (R): The perpendicular bisector of OP helps in finding the midpoint which acts as the center of a circle passing through O and P.

Answer: (A)

Solution:

1. To find points of contact, we draw a circle with diameter OP.
2. The midpoint is found by constructing the perpendicular bisector of OP.

Derivation: A: True, R: True, R explains A *Constructions — construction of tangents*

25. **25.** Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameter each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q.

Answer:

Two pairs of tangents are constructed from P and Q.

Solution:

1. Draw a circle of radius 3 cm with centre O. Extend the diameter on both sides to points P and Q such that $OP = OQ = 7$ cm.
2. Bisect OP at M and OQ at N. Draw circles with centers M and N and radius MP and NQ respectively, intersecting the original circle at the required points. Join the points of intersection to P and Q respectively.

Derivation: Construction involves perpendicular bisector method to find the midpoints of the lines connecting the external points to the center, creating circles that define the tangent contact points. *Constructions — construction of tangents*

26. **26.** Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.

Answer:

The length of the tangent is approximately4.47 cm.

Solution:

1. Draw two concentric circles with centre O and radii 4 cm and 6 cm. Let P be a point on the outer circle.
2. Join OP. Bisect OP at M. Draw a circle with centre M and radius MO to intersect the inner circle at A and B. Join PA and PB, which are the required tangents.

Derivation:

$$L = \sqrt{R^2 - r^2} = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} \approx 4.47 \text{ cm.}$$

Constructions — construction of tangents

27. **27.** Draw a line segment AB of length 8 cm. Taking A as centre, draw a circle of radius 4 cm and taking B as centre, draw another circle of radius 3 cm. Construct tangents to each circle from the centre of the other circle.

Answer:

Tangents constructed from B to circle A and from A to circle B.

Solution:

1. Draw segment $AB = 8$ cm. Construct circles of radii 4 cm and 3 cm at A and B.
2. Bisect AB at M. Draw a circle with centre M and radius MA (or MB). This circle intersects the circle at A at two points and the circle at B at two points. Join these points to B and A respectively.

Derivation: The construction uses the property that the angle in a semicircle is 90ˆ∘ , thus defining the points of tangency. *Constructions — construction of tangents*

28. **28.** Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60°.

Answer:

The tangents are constructed such that the angle between radii at the point of contact is120°.

Solution:

1. Draw two radii OA and OB such that $\angle AOB = 120^\circ$ (since $180^\circ - 60^\circ = 120^\circ$).
2. Construct perpendiculars at A and B to radii OA and OB. These perpendiculars meet at point P, where $\angle APB = 60^\circ$.

Derivation: The quadrilateral formed by the center, the two points of contact, and the external point has angles summing to 360ˆ∘ . Since the radii are perpendicular to tangents, ∠ OAP = ∠ OBP = 90ˆ∘ . *Constructions — construction of tangents*

29. **29.** Draw a circle of radius 5 cm. Mark a point P at a distance of 8 cm from the centre. Construct the two tangents from P to the circle.

Answer:

Two tangents from P are constructed.

Solution:

1. Draw a circle with centre O and radius 5 cm. Mark point P such that $OP = 8$ cm.
2. Bisect OP at M. Draw a circle with centre M and radius MO. Mark the intersection points A and B on the original circle. Join PA and PB.

Derivation: The construction relies on the property that the tangent is perpendicular to the radius at the point of contact, creating a right-angled triangle OAP . *Constructions — construction of tangents*

30. **30.** Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q.

Answer:

$$\text{Two pairs of tangents constructed from points at distance} 7 \text{ cm}$$

Solution:

1. Draw a circle with center O and radius 3 cm.
2. Draw a diameter and extend it on both sides to locate points P and Q such that $OP = OQ = 7$ cm.
3. Bisect OP and OQ to find midpoints M_1 and M_2.
4. With M_1 and M_2 as centers, draw circles of radius M_1O to intersect the original circle at two points each. Join P and Q to these intersection points.

Derivation: Construction follows the geometric property: angle in a semicircle is 90ˆ○ . *Constructions — Construction of tangents*

31. **31.** Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.

Answer:

$$\sqrt{20} \approx 4.47 \text{ cm}$$

Solution:

1. Draw two concentric circles with radii 4 cm and 6 cm with center O.
2. Mark a point P on the circumference of the larger circle.
3. Join OP and bisect it to find midpoint M.
4. Draw a circle with center M and radius MO to cut the inner circle at A and B. PA and PB are the required tangents.

Derivation:

$$\text{Length L} = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} = 2\sqrt{5}$$

Constructions — Construction of tangents

32. **32.** Draw a line segment AB of length 8 cm. Taking A as center, draw a circle of radius 4 cm and taking B as center, draw another circle of radius 3 cm. Construct tangents to each circle from the center of the other circle.

Answer:

$$\text{Two pairs of tangents constructed}$$

Solution:

1. Draw segment $AB = 8$ cm. Draw circle C_1 (center A, $r = 4$) and C_2 (center B, $r = 3$).
2. Bisect AB to get midpoint M.
3. Draw a circle with center M and radius MA (or MB).

4. The points where this circle intersects C_1 and C_2 are the points of tangency. Join B to intersection points on C_1 and A to intersection points on C_2.

Derivation: Tangents are constructed by finding the intersection of circles centered at the midpoint of the line segment connecting the centers. *Constructions — Construction of tangents*

33. 33. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60°.

Answer:

$$\text{Construction of tangents at} 120^\circ \text{central angle}$$

Solution:

1. Draw a circle of radius 3.5 cm with center O.
2. Draw a radius OA. Draw another radius OB such that $\angle AOB = 120^\circ$ (since $180^\circ - 60^\circ = 120^\circ$).
3. At point A, draw a perpendicular to OA.
4. At point B, draw a perpendicular to OB. The intersection point P gives the required tangents PA and PB.

Derivation: Sum of opposite angles in the quadrilateral OAPB is 180ˆ∘ . Thus, if ∠ APB = 60ˆ∘ , ∠ AOB = 120ˆ∘ . *Constructions — Construction of tangents*

34. 34. Construct a triangle with sides 5 cm, 6 cm and 7 cm. Then construct a circle touching the side of 6 cm at its midpoint and passing through the other vertex.

Answer:

$$\text{Construction of a circle through specific vertices tangent to a side}$$

Solution:

1. Construct $\triangle ABC$ with $AB = 5, BC = 6, AC = 7$.
2. Locate midpoint M of BC.
3. Draw a perpendicular to BC at M.
4. Draw the perpendicular bisector of AM. The intersection of these two perpendiculars is the center O. Draw the circle with radius OM.

Derivation: The center must be equidistant from the point of tangency M and vertices A and C . *Constructions — Construction of tangents*

35. 35. Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q.

Answer:

$$\text{Two pairs of tangents constructed from points P and Q, each pair having equal lengths of} \sqrt{7^2 - 3^2} = \sqrt{40} \approx 6.32 \text{ cm.}$$

Solution:

1. Draw a circle with center O and radius 3 cm.
2. Draw a diameter and extend it on both sides to locate P and Q such that OP = OQ = 7 cm.
3. Bisect OP and OQ to find midpoints M and N.
4. Using M as center and MO as radius, draw a circle to intersect the original circle at points A and B.
5. Using N as center and NO as radius, draw a circle to intersect the original circle at points C and D.
6. Join PA, PB, QC, and QD to get the required tangents.

Derivation: Length of tangent = $\sqrt{d^2 - r^2} = \sqrt{7^2 - 3^2} = \sqrt{49 - 9} = \sqrt{40}$ cm . *Constructions — Construction of tangents to a circle*

36. 36. Construct a triangle ABC with side $BC = 6$ cm, $AB = 5$ cm and $\angle ABC = 60°$. Then construct a triangle whose sides are $\frac{3}{4}$ of the corresponding sides of the triangle ABC. Finally, construct a tangent to the circumcircle of triangle ABC at vertex A.

Answer:

Ascaled triangle A$'BC'$and a tangent line touching the circumcircle at A.

Solution:

1. Construct triangle ABC with given dimensions.
2. Draw an acute angle at B and divide the line into 4 segments to construct the similar triangle A'BC'.
3. Find the circumcenter O by drawing perpendicular bisectors of AB and BC.
4. Draw the circumcircle with center O passing through A, B, and C.
5. Construct a line perpendicular to radius OA at point A to get the tangent.

Derivation: The tangent at A is perpendicular to the radius OA, hence ∠ OAT = 90ˆ∘ . *Constructions — Construction of tangents and similar triangles*

37. 37. Draw a pair of tangents to a circle of radius 4 cm, which are inclined to each other at an angle of 60°. Justify the construction.

Answer:

Tangents constructed such that the angle between them is60degrees.

Solution:

1. Draw a circle of radius 4 cm with center O.
2. Draw a radius OA.
3. Draw a radius OB such that $\angle AOB = 180° - 60° = 120°$.
4. At point A, construct a line perpendicular to OA.
5. At point B, construct a line perpendicular to OB.

6. The point where these two lines intersect is P. $\angle APB$ will be 60°.

Derivation: In quadrilateral OAPB, ∠ OAP = 90ˆ∘ , ∠ OBP = 90ˆ∘ , and ∠ AOB = 120ˆ∘ . Thus ∠ APB = 360ˆ∘ − (90° + 90° + 120°) = 60° . *Constructions — Construction of tangents*

38. 38. Draw two concentric circles of radii 3 cm and 5 cm. Construct a tangent to the smaller circle from a point on the larger circle and measure its length. Verify the measurement by calculation.

Answer:

$$\text{Length of tangent is} 4cm.$$

Solution:

1. Draw two concentric circles with center O and radii 3 cm and 5 cm.
2. Mark a point P on the circumference of the larger circle.
3. Join OP (length 5 cm).
4. Bisect OP at M.
5. Draw a circle with center M and radius MO, cutting the smaller circle at A and B.
6. Join PA and PB, which are the required tangents.

Derivation: In right △ OAP , OA = 3 cm , OP = 5 cm . By Pythagoras theorem, PA = $\sqrt{OP^2 - OA^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4$ cm . *Constructions — Construction of tangents*

39. 39. If the radius of a sphere is 7 cm, then its surface area is:

Answer: (C)

Solution:

1. State the formula for the surface area of a sphere: $A = 4\pi r^2$.
2. Substitute $r = 7$ cm and $\pi = \frac{22}{7}$.
3. Calculate the final value.

Derivation:

$$A = 4 \times \frac{22}{7} \times 7 \times 7 = 4 \times 22 \times 7 = 616 \text{ cm}^2$$

Surface Areas and Volumes — Surface area of sphere

40. 40. The volume of a right circular cylinder with radius 3 cm and height 7 cm is:

Answer: (B)

Solution:

1. State the formula for the volume of a cylinder: $V = \pi r^2 h$.
2. Substitute $r = 3$ cm, $h = 7$ cm, and $\pi = \frac{22}{7}$.

3. Calculate the final value.

Derivation:

$$V = \frac{22}{7} \times 3^2 \times 7 = 22 \times 9 = 198 \text{ cm}^3$$

Surface Areas and Volumes — Volume of cylinder